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INTERIM REVISION OF
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INTERIM FEDERAL SPECIFICATION

BOTTLE, WEIGHING (WITH STOPPER)

This Interim Federal Specification was developed by the General Services Administration, Federal Supply Service, Standardization Division, Washington, D. C. 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

The General Services Administration has authorized the use of this Interim Federal Specification as a valid exception to Federal Specification DD-B-620A, dated November 14, 1957.

1. SCOPE AND CLASSIFICATION

1.1 Scope. The items covered by this specification shall be weighing bottle with stoppers, for laboratory use.

1.2 Classification. Weighing bottles covered by this specification shall be of the following types, styles and sizes, as specified (see 6.2).

1.2.1 Types. Types shall be applicable to all styles and sizes.

(a) Type I - borosilicate glass.
(b) Type II - soda-lime glass.

1.2.2 Styles and sizes. Sizes, as shown, shall be the dimensions of the bottles-inside diameter x height, in millimeters (mm.).

(a) Style I - Weighing bottles with plug style stopper.

Size 1	- 10 x 17
Size 2	- 15 x 50
Size 3	- 15 x 80
Size 4	- 25 x 40
Size 5	- 25 x 50
Size 6	- 30 x 50
Size 7	- 30 x 60
Size 8	- 40 x 50
Size 9	- 40 x 80
Size 10	- 40 x 100
Size 11	- 45 x 65
Size 12	- 50 x 30
Size 13	- 50 x 60
Size 14	- 60 x 30
Size 15	- 70 x 30
Size 16	- 70 x 33

(b) Style II - Weighing bottles with cap style stoppers.

Size 1	- 7.5 x 15
Size 2	- 15 x 50
Size 3	- 15 x 80
Size 4	- 20 x 24
Size 5	- 25 x 20
Size 6	- 25 x 40
Size 7	- 25 x 50
Size 8	- 30 x 60
Size 9	- 40 x 50
Size 10	- 40 x 80
Size 11	- 40 x 100
Size 12	- 50 x 30
Size 13	- 60 x 30
Size 14	- 70 x 33

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2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

Federal Specifications:

DD-G-541 - Glass (Laboratory).
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Fiberboard.

Federal Standard:

Fed. Test Method Std. No. 626/401 - Glassware: Methods for Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-L-10547 - Liners, Case, and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

C 148 - Polariscopic Examination of Glass Containers.
C 149 - Thermal Shock Test on Glass Containers.
C 162 - Terms Relating to Glass and Glass Products.

(Application for copies should be addressed to the American Society for Testing and Material, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

U. S. Department of Commerce (CS) Standard:

CS 21 - Interchangeable Taper-Ground Joints, Stopcocks, Stoppers, and Spherical-Ground Joints.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.)

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago, Illinois 60606.)

National Motor Freight Classification Rules.

(Application for copies should be addressed to the National Classification Board, 1616 P Street, N. W., Washington, D. C. 20036.)

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(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Type I bottles. Glass used in the manufacture of type I weighing bottles shall conform to the requirements of DD-G-541 for type I, class A borosilicate glass.

3.1.2 Type II bottles. Glass used in the manufacture of type II weighing bottles shall conform to the requirements of DD-G-541 for type II, soda-lime glass.

3.2 Design.

3.2.1 Bottles. Weighing bottles shall be cylindrical in shape, with a flat base which shall afford stable equilibrium. Body shall be the same diameter up to the base of the neck.

3.2.1.1 Style I bottles. In addition to 3.2.1, style I weighing bottles shall have a standard taper (%) glass neck ground to receive an inside fitting, plug-style stopper. Standard taper necks shall conform to CS 21.

3.2.1.2 Style II bottles. In addition to 3.2.1, style II weighing bottles shall have a standard taper (%) glass neck ground to receive an outside-fitting, cap style stopper.

3.2.2 Stoppers.

3.2.2.1 Style I stoppers. Style I glass stoppers shall be standard taper glass stopper conforming to CS 21. Stopper shall fit into the inside of the bottle neck and shall provide an airtight fit. Each stopper shall have a closed bottom and a vertical handle formed as an integral part of the stopper.

3.2.2.2 Style II stoppers. Style II glass stoppers shall be standard taper glass stoppers conforming to CS 21. Stopper shall fit onto the outside of the neck of the bottle and shall provide an airtight fit.

3.3 Capacity point. The capacity point of the bottle shall fall below the lower level of the ground neck.

3.4 Thermal shock.

3.4.1 Type I bottles. Type I weighing bottles shall not crack, chip, or break when tested as specified in 4.5.1.1.

3.4.2 Type II bottles. Type II weighing bottles shall not crack, chip, or break when tested as specified in 4.5.1.2.

3.5 Annealing. Ring strain shall not be greater than that of temper no. 1 when tested as specified in 4.5.2. Longitudinal strain, if present, shall be faint and highly diffused.

3.6 Dimensions. The limiting dimensions of the weighing bottles shall be as shown in table I.

3.6.1 Wall thickness. The thickness of the walls of the bottle shall be uniform and maintained as light as practical consistent with the size of the bottle.

3.7 Markings. All markings shall be permanent and legible and shall not change in appearance when tested for permanency of marking in 4.5.3.

3.7.1 Identification marking. Each bottle shall be marked with the name or trademark of the manufacturer. Corresponding identification number and standard taper sizes shall be etched upon both the bottle and the stopper.

3.7.2 Marking area. Each bottle shall have an area on one side roughened by sandblasting or suitably treated to provide for marking with pencil, either serial numbers or inscription. The area shall show no removal of enamel (except for dulling) and shall be satisfactory for marking when subjected to tests specified in 4.5.3.

3.8 Workmanship. The quality of the workmanship shall be first class throughout. All weighing bottles shall be transparent, properly finished, and free from any defects which detract from their appearance or may impair their serviceability. Bottles shall be symmetrical in design and the glass shall be uniformly distributed.

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TABLE I. Dimensional requirements

Classification			Inside diameter	Height	Approximate capacity	Standard taper joint
Style	Type	Size	Millimeter	Millimeter	Milliliter	¶
I	I,II	1	10	17	1	10/7
I	I,II	2	15	50	7	14/10
I	I,II	3	15	80	12	14/10
I	I,II	4	25	40	12	24/12
I	I,II	5	25	50	16	24/12
I	I,II	6	30	50	25	29/12
I	I,II	7	30	60	30	29/12
I	I,II	8	40	50	45	40/12
I	I,II	9	40	80	70	40/12
I	I,II	10	40	100	95	40/12
I	I,II	11	45	65	60	45/12
I	I,II	12	50	30	35	50/12
I	I,II	13	50	60	85	50/12
I	I,II	14	60	30	50	60/12
I	I,II	15	70	30	60	71/15
I	I,II	16	70	33	82	71/15
II	I,II	1	7.5	15	0.6	10/6
II	I,II	2	15	50	7	19/10
II	I,II	3	15	80	12	19/10
II	I,II	4	20	24	4	24/12
II	I,II	5	25	20	12	29/12
II	I,II	6	25	40	12	29/12
II	I,II	7	25	50	16	29/12
II	I,II	8	30	60	30	34/12
II	I,II	9	40	50	45	45/12
II	I,II	10	40	80	70	45/12
II	I,II	11	40	100	92	45/12
II	I,II	12	50	30	36	55/12
II	I,II	13	60	30	42	60/12
II	I,II	14	70	33	82	71/15

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 for which specific inspection has not been provided has been satisfied.

4.2 Lotting. A lot shall consist of all weighing bottles of the same type, style, and size, produced by one manufacturer for delivery at one time.

4.3 Sampling.

4.3.1 Sampling for examination. Sampling for nondestructive examination of visual and dimensional characteristics of the end item shall be conducted in accordance with MIL-STD-105. Unit for sampling shall consist of one weighing bottle with stopper.

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	Inspection Level	AQL (percent defective)
For visual examination		
Major defects	II	2.5
Minor defects	II	4.0
For dimensional examination	S-3	2.5

4.3.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 (see table II). A sampling of five bottles and stoppers shall be taken at random from each 100 bottles and stoppers or fraction thereof in the lot (see 4.2), for tests.

TABLE II. Testing requirements

Characteristic	Requirement	Test procedure	Inspection level	AQL (defects per 100 units)
Thermal shock	3.4		S-2	*
Type I		4.5.1.2		
Type II		4.5.1.2		
Annealing	3.5	4.5.2	S-2	*
Permanency of marking	3.7	4.5.3	S-2	*

*Acceptance number shall be zero.

4.4 Examinations.

4.4.1 Visual examinations. The bottles shall be examined (see 4.1.1), to determine compliance with the applicable requirements of this specification. Examination shall be conducted in accordance with the classification of defects below. Definition of glass defects shall be in accordance with ASTM C 162. Examination shall not be restricted to the defects listed.

Classification of defects for visual examination

Major	Categories and defects
101	Glass in walls of bottle not uniformly distributed.
102	Bottom of bottles do not rest evenly upon flat, smooth surface.
103	Bottle and stopper not permanently and legibly marked with the name of manufacturer, standard taper size, and identification numbers.
104	Bottle and stopper not free of cracks, chips, or breaks.
105	Incorrect standard taper joints.
106	Standard taper joints not rated correctly.
107	Bottle and stopper not free of stones or blisters.
108	Bottle not free of cloudiness.
109	Bottle not free of grease, dirt, or other foreign material.
Minor	
201	Bottle and stopper not free of bubbles or striae.
202	Bottle not free of sharp or rough edges and protrusions.

4.4.2 Dimensional examination. Weighing bottles shall be examined for defects in dimensions. Any dimension not within the tolerance specified shall be classified as a major defect.

4.5 Tests.

4.5.1 Thermal shock.

4.5.1.1 Type I. Type I weighing bottles shall be tested for thermal shock in accordance with Fed. Test Method Std. No. 626, method 401.

4.5.1.2 Type II. Type II weighing bottles shall be tested for thermal shock in accordance with ASTM C 149.

4.5.2 Annealing. Weighing bottles shall be tested in accordance with method A or B of ASTM C 148.

4.5.3 Permanency of marking. A different weighing bottle shall be used for each of the following tests.

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4.5.3.1 Resistance to alkali. The weighing bottle shall be completely immersed in concentrated 1N sodium hydroxide solution at room temperature, covered, boiled for one-half hour, and allowed to stand in the alkali for an additional hour. The bottle shall then be removed, thoroughly rinsed with distilled water, and dried by rubbing vigorously with a cloth.

4.5.3.2 Resistance to acid.

4.5.3.2.1 Hydrochloric acid test. The weighing bottle shall be completely immersed in concentrated hydrochloric acid (specific gravity 1.18), at room temperature, covered, boiled for one-half hour, and allowed to stand in the acid for an additional one hour. The bottle shall then be removed, thoroughly rinsed with distilled water, and dried by rubbing vigorously with a cloth.

4.5.3.2.2 Sulfuric acid test. The bottles shall be completely immersed in concentrated sulfuric acid (specific gravity 1.84), at room temperature and allowed to stand for one hour. The bottle shall then be removed, thoroughly rinsed with distilled water, and dried by rubbing vigorously with a cloth.

4.6 Examination of preparation for delivery. An examination shall be made to determine compliance with the packaging, packing, and marking requirements in section 5. Defects shall be scored in accordance with table III. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be sealed. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the acceptable quality level shall be 4.0 defects per hundred units.

TABLE III. Preparation for delivery defects

Examine	Defects
Packaging	Unit packaging not as specified.
Material	Any nonconforming component, component missing, damaged, or otherwise defective.
Containers	Not as specified.
Contents	Quantity in interior packages and shipping containers not as specified.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, container bulged or distorted.
Markings	Omitted, incorrect; illegible; improper size, location sequence or method of application.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One bottle with stopper, as specified, constitutes one unit.

5.1.2 Quantities. Packaging and packing quantities shall be as specified in table IV.

TABLE IV. Packaging and packing quantities

Unit pack	Intermediate pack	Shipping container
Not required	12 units	144 units

5.1.2.1 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit preparation. Stopper shall be inserted into the bottle with a strip of paper interposed.

5.1.3.2 Intermediate package. The number of units specified in table IV shall be packaged in a carton of appropriate size constructed in accordance with PPP-B-621 or PPP-B-636. Carton shall contain double-faced corrugated fiberboard partitions and pads, or other suitable cushioning materials to insure against shifting or breakage of contents. Carton shall be closed in accordance with requirements of the applicable specification.

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5.1.4 Level C. Weighing bottles shall be packaged to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with Uniform Freight Classification Rules or National Motor Freight Classification Rules or other common carrier regulations applicable to the mode of transportation.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. The number of units specified in table IV shall be overpacked in an exterior container conforming to class 2 of PPP-B-621 and strapped in accordance with appendix I thereof except that strapping will not be required for shipment forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.1.2 Case liner. Each box shall be lined with a waterproof case liner conforming to type I, grade A or B, class 2 of MIL-L-10547. Closure and sealing shall be in accordance with applicable paragraphs of appendix thereof.

5.2.2 Level B. The number of units specified in table IV shall be overpacked in an exterior container conforming to class 1 of PPP-B-621 or PPP-B-636, class domestic.

5.2.3 Level C. The weighing bottles, packaged as specified in 5.1.4, shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with the Uniform Freight Classification Rules or National Motor Freight Classification Rules or other common carrier regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 Civil agencies. Unless otherwise specified, issue packages and shipping containers shall be marked with the name and class of item, stock number, the quantity contained therein, the name of the contractor, and the number of the contract or order.

5.3.2 Military agencies. In addition to any special marking required by the contract or order, all shipments shall be marked in accordance with MIL-STD-129. All shipping containers shall be conspicuously marked "Glass-Handle with care". Service color marking and symbol are not required on container for level A packs when addressed to domestic establishments for shipments for the Medical Services of the Armed Forces.

6. NOTES

6.1 Intended use. Weighing bottles covered by this specification shall be used to hold samples for the purpose of weighing.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Level of packaging and packing required (see 5.1 and 5.2).
(c) Type, style, and size of weighing bottle required (see 1.2).
(d) Special marking required (see 5.3.1 and 5.3.2).

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